NCSN Cheat Sheet

1. The Forward Process (Noise Injection)

The Formula:

$$\tilde{x} = x + \sigma\epsilon$$

(Alternatively written as $\tilde{x} = x + \epsilon$ where $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$)

Variables:

- x: The clean, original data point.
- σ: The current noise level (standard deviation).
- ϵ: The sampled standard normal noise vector $\mathcal{N}(0, I)$.
- $\tilde{x}$: The resulting noise-perturbed data point.

When to use it:

- "Calculate the input to the Noise-Conditional Score Network."
- "Find the coordinates of the corrupted/perturbed image."
- "Given a clean point x and a sampled Gaussian noise vector, determine the state of the system before the network makes a prediction."

2. The True Score of the Gaussian (The Target)

The Formula:

$$S_{true}(\tilde{x}|x) = -\frac{\epsilon}{\sigma}$$

(Note: If the noise ϵ was defined with variance σ^2 built-in during the problem setup, this is written as $-\frac{\epsilon}{\sigma^2}$ or $\frac{x-\tilde{x}}{\sigma^2}$)

Variables:

- ϵ: The exact noise vector that was added to the image.
- σ^2: The variance of the noise level.

When to use it:

- "Calculate the ground-truth score of the conditional distribution."

- "What is the theoretical target vector the network is trying to predict?"
- "Compute $\nabla_{\tilde{x}} \log P_\sigma(\tilde{x}|x)$ for this specific step."
- "What is the exact gradient direction pointing back to the clean data?"

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3. The Unweighted NCSN Loss

The Formula:

$$Loss = \frac{1}{2} \left\| S_\theta(\tilde{x}, \sigma) + \frac{\epsilon}{\sigma^2} \right\|_2^2$$

Variables:

- $S_\theta(\tilde{x}, \sigma)$: The vector predicted by your neural network.
- $\frac{\epsilon}{\sigma^2}$: The true score (target).

When to use it:

- "Calculate the raw Denoising Score Matching loss."
- "Compute the unweighted squared error for the network's prediction."
- "Find the basic loss before applying any noise-level balancing factors."
- *Trap Warning:* If the question does *not* mention $\lambda(\sigma)$ or weighting, default to this basic Euclidean distance.

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4. The Weighted NCSN Objective (With λ)

The Formula:

$$L(\theta) = \|\sigma S_\theta(\tilde{x}, \sigma) + \epsilon\|_2^2$$

Variables:

- This is the simplified version of the loss assuming the professor's weighting factor is $\lambda(\sigma) = \sigma^2$.

When to use it:

- "Calculate the weighted loss for this training step."
- "Compute the final objective $L(\theta)$ assuming $\lambda(\sigma) = \sigma^2$."
- "What is the noise-balanced error for this prediction?"

- *Trap Warning:* Ensure you multiply the network's prediction by σ *before* adding ϵ and squaring the result.

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5. ALD Step Size Calculation (α_i)

The Formula:

$$\alpha_i = \epsilon_{step} \cdot \frac{\sigma_i^2}{\sigma_L^2}$$

Variables:

- ϵ_{step}: The base step-size hyperparameter (often just written as ϵ in the algorithm pseudo-code, but do not confuse it with the noise vector).
- σ_i^2: The variance of the *current* noise level you are stepping through.
- σ_L^2: The variance of the *smallest* (final) noise level in the entire schedule.

When to use it:

- "Calculate the step size α for noise level i."
- "Determine the scaling factor for the Langevin update at this specific schedule step."
- "Before applying the gradient, find the dynamic learning rate/step magnitude."

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6. Annealed Langevin Dynamics (ALD) Update Rule

The Formula:

$$\tilde{x}_t = \tilde{x}_{t-1} + \frac{\alpha_i}{2} S_\theta(\tilde{x}_{t-1}, \sigma_i) + \sqrt{\alpha_i} Z_t$$

Variables:

- $\tilde{x}_{t-1}$: Your current image coordinates.
- α_i: The step size computed in Formula 5.
- $S_\theta(\tilde{x}_{t-1}, \sigma_i)$: The score vector predicted by the network.
- Z_t: A fresh random noise vector drawn from $\mathcal{N}(0, I)$ just for this specific step.

When to use it:

- "Perform one step of Annealed Langevin Dynamics."
- "Calculate the next image state $\tilde{x}_t$."

- "Apply the inner-loop corrector step."
- "Given the network's prediction and the injected noise Z_t, find the updated coordinates."
- *Trap Warning:* Remember to divide the step size by 2 for the gradient term, but take the square root of the step size for the noise term.

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7. The DDPM-to-NCSN Score Conversion

The Formula:

$$S_\theta(x_t, t) = -\frac{\epsilon_\theta(x_t, t)}{\sqrt{1-\bar{\alpha}_t}}$$

Variables:

- $\epsilon_\theta(x_t, t)$: The noise vector predicted by a DDPM network.
- $\bar{\alpha}_t$: The cumulative variance schedule product from DDPM at step t.
- $S_\theta(x_t, t)$: The equivalent Score vector.

When to use it:

- "Given this DDPM noise prediction, what is the equivalent score?"
- "Convert the predicted noise vector ϵ_θ to a score vector."
- "If we swap to a score-based sampler, what is the implicit score of this DDPM state?"
- "Calculate $\nabla_{x_t} \log q(x_t|x_0)$ based on the DDPM output."